

SEVAN DALZELL

Final Year Mechatronics Student

EXPERIENCE

- Industrial Engineering Intern

Elitepac NZ Ltd

Dec. 2024 – Jan. 2025

Returned Dec. 2025 – Mar. 2026

 - Designed and established a WorkSafe-compliant segregated hazardous substance storage system, & coordinated regulatory approvals and audits for WorkSafe & BASF SE
 - Set up key factory infrastructure (security systems, waste management, Wi-Fi, PPE/signage) & conducted market research for chemical formulation
- Research Assistant

The University of Auckland

Oct. 2024 – Dec. 2025

 - Conducted acoustic measurements in real-world environments (e.g. large halls), analysing frequency response characteristics of different spaces
 - Developed automated methods to synchronise misaligned acoustic datasets, restoring usability of experimental data
 - Acted as a technical liaison between research groups, coordinating data processing approaches and ensuring effective cross-team communication
- Engineering Tooling Assistant

Trade Tools Ltd

Jan. 2025 – Mar. 2025

 - Assisted with engineering supply procurement, managed inventory, and collaborated with multiple teams to support efficient throughput

VOLUNTEERSHIP & COMMUNITY INVOLVEMENT

- FoED & Mech. Staff-Student Consultative Committee

The University of Auckland

2025 – Present

 - Rainbow Engineering representative at faculty-wide staff meetings
- Video Director, Writing & Video Team Executive

The Engineering Revue

2024 – Present

 - Oversee full production workflow from concept to filming, for screening of two industry-standard short films at the Engineering Revue, coordinating operations, budgeting, scheduling and thorough interdepartmental collaboration
- General & Marketing Executive

Rainbow Engineering UoA

2024 – Present

 - Executive involvement in events and marketing planning for UoA's LGBTQIATakatāpui+ Engineering faculty. Oversaw a **64%** increase in peak-time sign-ups during marketing tenure

- Event Photography & Videography

Various Organisations

 - Event photographer for Hardware Meetup NZ, UoA's MECHA, AUDA, AUES, Rainbow Engineering, Engineering Revue, and Women in Engineering Network
- Volunteer

Know Your Stuff NZ

2022 – Present

 - Provide community drug-checking services for harm reduction and education, developing practical experience with FTIR spectroscopy

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TECHNICAL SKILLS

- Practical / Mechanical:** Data Processing & Analysis, MIG Welding, Turning & Milling, CAD, 3D printing, FTIR Spectroscopy, Photogrammetry
- Electrical:** Soldering, Circuit Design, Microcontrollers, Analog Sensor & Digital Signal Processing
- Documentation:** Technical Writing, SOP Writing, Technical drawing
- Coding Languages:** C, C++, MATLAB, R, $\text{\LaTeX}$
- Software:** Adobe Creative Cloud, Autodesk Inventor, Blender, Simulink, Git, Ansys

SOFT SKILLS

- Teamwork:** Team Cohesion, Conflict Resolution, Cross-Team Communication, Relationship Building, Collaboration, Leadership, Mentorship
- Problem Solving:** Initiative, Resilience, Analytical Thinking, Adaptability, Creativity, Proactive Learning
- Organisation:** Time Management, Organisation, Critical Thinking, Attention to Detail, Flexibility, Strong Drive for Excellence

EDUCATION

- BE(Hons) Mechatronics Engineering, Ongoing

The University of Auckland

2023 – Present

PROJECTS

- 2025 DTRG Indoor Drone Competition
 - Competed in a team drone navigation and payload mechanism design competition
- 2025 MECHA Design Challenge, 2nd Place
 - Designed and built an aquatic robot to specification in a one-day team engineering challenge
- Personal Project: RPi Wi-Fi Access Point
 - Set up a Raspberry Pi as a custom internet access point to connect older devices to University Wi-Fi
- University Project: 2-Axis CNC Controller
 - Developed in C++, a Ziegler-Nichols tuned PID controller for a G-code pencil plotter, achieving **0.012 mm** resolution via precise voltage control

LANGUAGES

English	Native
Japanese	Intermediate (~JLPT N5)

- REFERENCES

MORE ON REQUEST
- Dr Justine Hui

Senior Lecturer, Department of Mechanical & Mechatronics Engineering

The University of Auckland

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